Integration

Speed of change → Differentiation $\frac{d}{dx}$

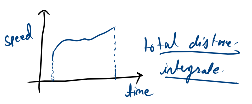
accumulated result

area under the curve.

inverse of differentiation

Day 1 → 10
Day 2 → 20
Day 3 → 10

after 3 days → 30



total distance.
integrate.

S → sum

$\int f(x)\,dx$

speed-time → distance.
power-time → energy
probability curve → total probability
salary-time → total earnings.

Power rule: → $\int x^n dx = x^{n+1}/(n+1) + C$

$\int x^2 dx = \frac{x^{2+1}}{2+1} + c \Rightarrow \frac{x^3}{3} + c$

$\frac{d(x^2+3)}{dx} = 2x$

$\int 2x\,dx = \frac{2x^{1+1}}{2} = x^2 + C$

Constant rule: →

$\int k\,dx = kx + c$

$\int 8\,dx = 8x + c$

$\int 9\,dy = 9y + c$

Sum/Diff rule: → $\int [f(x) \pm g(x)]\,dx = \int f(x)\,dx \pm \int g(x)\,dx$

$\int (x^2+3x)\,dx = \int x^2 dx \pm \int 3x\,dx + c$

$= \frac{x^{2+1}}{2+1} \pm 3\frac{x^{1+1}}{1+1} + c$

$= \frac{x^3}{3} + \frac{3x^2}{2} + c$

exponential rule: → $\int e^x dx = e^x + C$

Natural log: → $\int \left(\frac{1}{x}\right) dx = \ln|x| + c$

① Definite integrals

$\int_{\text{lower limit}}^{\text{upper limit}}$ → first integrate → substitute the limits.

$3\times3\times3\times3$

$\int_1^3 x^3 dx$

$\int x^n dx = \frac{x^{n+1}}{n+1}$

$\rightarrow \frac{x^{3+1}}{3+1} = \left[\frac{x^4}{4}\right]_1^3 \Rightarrow \frac{3^4}{4} - \frac{1^4}{4}$

$\Rightarrow \frac{81}{4} - \frac{1}{4} \Rightarrow \frac{80}{4} = \boxed{20}$

Indefinite integrals: →

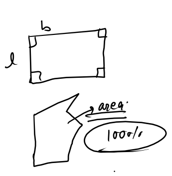
$\int_0^3 x^2 dx \Rightarrow \frac{x^{2+1}}{2+1} \Rightarrow \frac{3^3}{3} - 0$

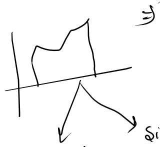
$\Rightarrow \frac{27}{3} = 9.$

$\int x^2 dx \Rightarrow \frac{x^3}{3} \Rightarrow \frac{x^{2+1}}{2+1} = \frac{x^3}{3}$

$\int_0^3 x^2 dx \Rightarrow \left[\frac{x^3}{3}\right]_0^3 \Rightarrow \left[\frac{3^3}{3} - \frac{0^3}{3}\right]$

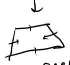
$\Rightarrow 9$





Simpson's rule

Trapezoid rule



area $= \frac{1}{2}(a+b)h$

x^2

x	$f(x)$
0	0
1	1
2	4

→ one trapezoid

$\int_a^b f(x)\,dx \approx \sum \frac{f(x_i) + f(x_{i+1})}{2} \Delta x$

left height → $f(x_i)$

right height → $f(x_{i+1})$

width $= \Delta x$

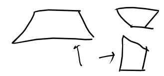
Average height $\Rightarrow \frac{f(x_i) + f(x_{i+1})}{2}$ Δx → trapezoids.

Simpson → parabolas → curves are curved

1 4 2 4 2 1

$\int_a^b f(x)\,dx \approx \frac{\Delta x}{3}\left[f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + 2f(x_4) + \dots f(x_n)\right]$

Trapezoid



Probability → curve

connection

event → happen / not happen

H, T

$\frac{1}{2}, \frac{1}{2}$

$\frac{1}{2} + \frac{1}{2} \rightarrow 1$

pos. of getting $= \frac{1}{6}$

LUDO. → 6, 1, 2, 3 $= \frac{1}{6}$

$\frac{2}{6} = x$

Normal distribution. → Certainity $= 1$

area under PDF $= 1$

$\boxed{\text{Stats. norm. pdf}}$

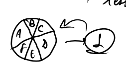
probability density function

$[-\infty, +\infty]$ PDF$(x)\,dx = 1$

PDF → How probability is distributed across different values. → curve

water → table → places have more water → higher probability
→ places it is less → lesser probability.

$\int_{-\infty}^{\infty} \text{PDF}(x)\,dx = 1$



μ → myu
→ mu